

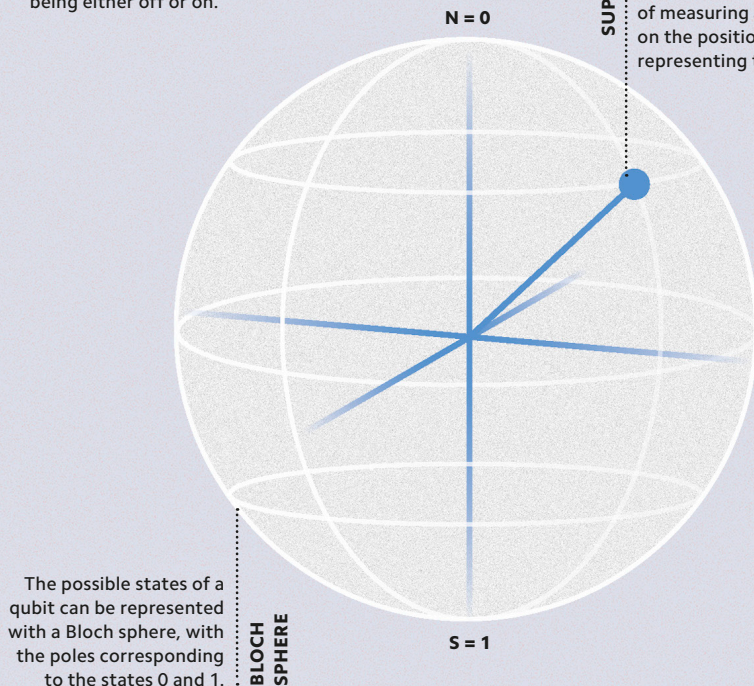
BIT COMPARISON

In a classical computer, a bit is either 0 or 1, which can also be thought of as a switch being either off or on.

0 1
BIT

SUPERPOSITION

A qubit can exist in a superposition of the states 0 and 1, represented by an arrow pointing at a location on a Bloch sphere. The probability of measuring a 0 or a 1 depends on the position of the vector representing the qubit.



NOT JUST ZEROS AND ONES

The basic unit of information in classical computing is the bit, which can take one of two states (represented as 0 or 1). The quantum equivalent, known as a “qubit,” is not in one state or the other, but instead in a superposition (see pp.38–39) of these two states. Qubits can be encoded in the properties of particles such as electron spin (up or down) and photon polarization (vertical or horizontal).